

Example Exam for Pre-requisite Reading Course Linear Models

Remarks:

You have **60** minutes. Provide **sufficient reasons** for your solutions! If you need additional pages, please enter your **name** and **student ID** on them, as well as the **number of the task**!

Allowed tools:

- non-programmable pocket calculator,
- a dictionary, if necessary.

Please check whether your exam sheet is complete. The exam should contain 2 problems on ?? numbered pages.

Round off all your intermediate results to 3 decimal places!

Please fill in legibly:

Name, Surname:	
Student-ID number:	
Study program:	

Is this your final attempt of this exam? Yes ☐ No ☐

Please leave **EMPTY**:

Exercise	1	2	Sum
possible Points	35	25	60
Points			

Note:

Exercies 1: (35 Points)

The football enthusiastic student Ms. Mossi is interested in the connection between the final points and the goals scored by the 18 teams that played in the German Bundesliga in the 2011/12 season. From the final table she takes that the teams scored an average of 48.61 goals and thereby have collected an average of 46.61 points.

Now she calculates a regression model in R and prints the output. However, when meeting her friend, Ms. Ronalda, she notes that there are missing parts in the R output:

```
Call: lm(formula = points ~ goals, data = bundesliga)
```

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	A	4.81560	C	0.595
Goals	0.90509	B	9.562	D

Multiple R-squared: 0.8511, Adjusted R-squared: 0.8418

F-statistic: 91.43 on 1 and 16 DF, p-value: 5.103e-08

- a) Calculate the missing values that should be in the gaps marked as A, B and C in the output. [4+4+4 P.]

Answer:

A: is rather easy. From 48.61 average goals and the corresponding estimated coefficient of 0.90509 we calculate the average influence of goals as $48.61 \cdot 0.90509 = 43.996$. The difference to $\bar{y} = 46.61$ is then the intercept:

$$46.61 - 43.996 = 2.614 = A$$

B: We know that $t_j = \frac{\hat{\beta}_j}{se_j}$ and can calculate

$$9.562 = \frac{0.90509}{se} \Leftrightarrow se = \frac{0.90509}{9.562} = 0.095$$

C: Same as B. Use $t_j = \frac{\hat{\beta}_j}{se_j}$:

$$C = \frac{2.614}{4.8156} = 0.543$$

- b) Explain how to get the entry of the gap marked with D. Pay particular attention to the required distribution including its parameters. [6 P.]

Answer: The t value corresponding to `goals` is 9.562. Under $H_0 : \beta_1 = 0$ it holds that $t_j = \frac{\hat{\beta}_j}{se_j} \sim t_{n-p-1}$. Here, with $n = 18$ and $p = 1$ this leads to the t -distribution with 16 degrees of freedom. The value D or $\Pr(>|t|)$ is now the quantile of that distribution corresponding to 9.562. It can be obtained via $(1 - \text{pt}(q = 9.562, df = 16)) * 2$ in R. The factor 2 is used to account for the absolute value being used and the t -distribution being symmetrical, such that both tails are accounted for.

For t_j to follow the t -distribution, β (and therefore β_j) is assumed to follow a Gaussian distribution.

Ms. Ronalda expresses fear that the estimate of the goals' influence is biased. She thereby refers to the result of two models, which she calculated in advance:

Output of Ms. Ronalda's first model:

```
Call: lm(formula = goals.received ~ goals, data = bundesliga)
```

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	77.0462	8.8376	8.718	1.79e-07
Goals	-0.5850	0.1737	-3.367	0.00392

Multiple R-squared: 0.4148, Adjusted R-squared: 0.3782
F-statistic: 11.34 on 1 and 16 DF, p-value: 0.003921

Output of Ms. Ronalda's second model:

```
Call: lm(formula = points ~ goals.received, data = bundesliga)
```

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	90.8270	7.3533	12.352	1.35e-09
Goals	-0.9096	0.1456	-6.245	1.17e-05

Multiple R-squared: 0.7091, Adjusted R-squared: 0.6909
F-statistic: 39 on 1 and 16 DF, p-value: 1.17e-05

c) Explain to what problem Ms. Ronalda is referring to with her fears. [4 P.]

Answer: She is worried about $\hat{\beta}$ being biased. This means that the expected value of the estimator is not the true parameter.

- unbiased: $E(\hat{\beta}) = \beta$
- biased: $E(\hat{\beta}) \neq \beta$

d) Is Ms. Ronalda right? Justify your answer! [6 P.]

Answer: She is worried about $\hat{\beta}$ being biased. This can be the case, if variables are missing. Here, the variable `goals.received` can be assumed to have an influence (see output of second model) and is missing in the main model. If the missing variable is not correlated with any variable in the model, the estimation remains unbiased. But here `goals` and `goals.received` are strongly correlated (see output of first model: `goals` has a significant influence on `goals.received`). So the main model's estimation can be expected to be biased.

e) Assume that $\beta_{goals.received} = -0.45763$ is the true parameter for the influence of the number of goals received on the number of points in a multiple regression model. How large is the bias of $\hat{\beta}_1$ in the model of Ms. Mossi? Calculate the coefficient $\hat{\beta}_0$ for the full model. [4 + 3 P.]

Answer: True effect of `goals.received` is -0.45763. From the output of the first model we learned that each scored goal reduced the number of goals received (not implying causality here) by 0.585. So a factor of $-0.585 \cdot (-0.45763) = 0.268$ needs to be adjusted into the main model. (every goal correlates to -0.585 more goals received and each goal received to -0.45763 more points).

$$\beta_{goals} = 0.90509 - 0.268 = 0.637$$

So the bias in $\beta_1 = \beta_{goals}$ is 0.268.

The average goals received are 48.61 as well (of course, every goal is counted as scored and received exactly once each).

$$\begin{aligned} & \beta_{goals} \cdot \overline{goals} + \beta_{goals.received} \cdot \overline{goals.received} \\ &= 0.637 \cdot 48.61 - 0.45763 \cdot 48.61 \\ &= (0.637 - 0.45763) \cdot 48.61 = 8.719 \end{aligned}$$

From 8.719 we are missing $46.61 - 8.719 = 37.891$ to the global average. So $\hat{\beta}_0 = 37.891$.

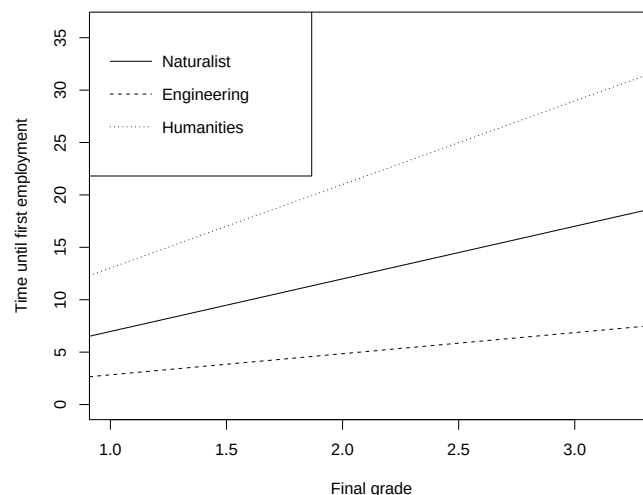
Exercies 2: (25 Points)

You now want to investigate whether the final grade X of the master thesis of three different groups of students (Group 1: naturalists, Group 2: engineers, Group 3: humanities) has an influence on the duration Y between the completion of the master thesis and the beginning of the first employment relationship (in weeks). For each of the three groups, you collected the final grade and the duration until the first employment from 10 students who are already in employment (rounded to two decimal places; for simplicity, this variable can be regarded to be interval scaled and continuous).

- Specify a suitable regression model **that includes interactions**. For the categorical variable that describes the subject studied, take reference encoding (i.e. dummy encoding), specifying Group 1 (natural scientists) as the reference category. To do this, also define explicitly suitable dummy variables and explicitly state the regression equation. Moreover, present also the design matrix X , which contains all the predictor variables. **[4 P.]**
- For each of the three groups, write down the general equation according to which the predicted durations can be calculated and combine the part describing the slope with the constant terms. How do the values of intercept and slope change for the three groups if effect encoding is used? **[6 P.]**
- Suppose you have obtained the following coefficient estimates:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	1.9505	0.2906	6.711	6.09e-07
as.factor(Group)2	-1.1278	0.3391	-3.326	0.00283
as.factor(Group)3	3.1035	0.3670	8.456	1.17e-08
X	5.0195	0.1064	47.163	< 2e-16
as.factor(Group)2:X	-3.0054	0.1279	-23.499	< 2e-16
as.factor(Group)3:X	2.9523	0.1314	22.467	< 2e-16

Explain how you can use these parameter estimates to obtain the coefficients of the regression line of the humanities group (see illustration). **[4 P.]**



- Interpret the parameters of the regression lines shown in the figure. **[6 P.]**
- On which group does final grade exhibits the largest influence? **[2 P.]**
- For all groups, calculate the exact estimated time to first employment for a final grade of 2.0 (without using the figure). **[3 P.]**

Solutions:

a) The regression equation is given by:

$$y_i = \beta_0 + \beta_1 x_i + \beta_2 \text{Group2}_i + \beta_3 \text{Group3}_i + \beta_4 \text{Group2} : x_i + \beta_5 \text{Group3} : x_i + \epsilon_i, \quad [1P.]$$

where $i = 1, \dots, n$ indexes the individuals / students in the sample and ϵ denotes the error term. The variable x denotes the final grade. Using dummy coding and setting naturalists as the reference category, **Group2** denotes whether a student has an engineering degree or not, i.e.

$$\text{Group2}_i = \begin{cases} 1, & \text{if student } i \text{ has engineering degree,} \\ 0, & \text{else.} \end{cases} \quad [0.5P.]$$

The variable **Group3** denotes whether a student has a humanities degree or not, i.e.

$$\text{Group3}_i = \begin{cases} 1, & \text{if student } i \text{ has humanities degree,} \\ 0, & \text{else.} \end{cases} \quad [0.5P.]$$

The remaining covariates **Group3:x** and **Group2:x** denote the interactions between the final grade x and the student-specific type of degree (engineering or humanities).

The design matrix of dimensions $n \times 6$ is given by: [2P.]

$$\begin{pmatrix} 1 & x_1 & 1 & 0 & x_1 & 0 \\ 1 & x_2 & 0 & 1 & 0 & x_2 \\ 1 & x_3 & 0 & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 1 & x_n & \text{Group2}_n & \text{Group3}_n & \text{Group2} : x_n & \text{Group3} : x_n \end{pmatrix}$$

(not sure if necessary: For example, here the first student has an engineering degree, the second has a humanities degree, and the third has a naturalist degree.)

b) The general equations are:

– Group 1: $E[y_i | \text{Group1}, X_i = x_i] = \hat{\beta}_0 + \hat{\beta}_1 \cdot x_i$ No combination of slope and intercept is required. [1.5P.]

– Group 2:

$$E[y_i | \text{Group2}, X_i = x_i] = \hat{\beta}_0 + \hat{\beta}_1 \cdot x_i + \hat{\beta}_{\text{Group2}} + \hat{\beta}_{\text{Group2} : x} \cdot x_i$$

$$\begin{aligned} \text{Here we combine: } \gamma_0 &= \hat{\beta}_0 + \hat{\beta}_{\text{Group2}}, \quad \gamma_1 = \hat{\beta}_1 + \hat{\beta}_{\text{Group2} : x} \\ &= \gamma_0 + \gamma_1 \cdot x_i \quad [1.5P.] \end{aligned}$$

– Group 3:

$$E[y_i | \text{Group3}, X_i = x_i] = \hat{\beta}_0 + \hat{\beta}_1 \cdot x_i + \hat{\beta}_{\text{Group3}} + \hat{\beta}_{\text{Group3} : x} \cdot x_i$$

$$\begin{aligned} \text{Here we combine: } \alpha_0 &= \hat{\beta}_0 + \hat{\beta}_{\text{Group3}}, \quad \alpha_1 = \hat{\beta}_1 + \hat{\beta}_{\text{Group3} : x} \\ &= \alpha_0 + \alpha_1 \cdot x_i \quad [1.5P.] \end{aligned}$$

Effect coding: The new effect coded variables would be:

$$\text{GroupJ}_i = \begin{cases} 1 & \text{if student } i \text{ has degree "J"} \\ -1 & \text{if student } i \text{ has naturalist degree} \\ 0 & \text{else} \end{cases}$$

The value of the model parameters using effect coding changes as follows: The intercept becomes the grand mean (average of the response across all groups), whereas the parameters for the group of students (i.e. $\hat{\beta}_{\text{Group2}}$, $\hat{\beta}_{\text{Group3}}$) become the differences between the respective group means, and the grand mean. [1.5P.] :

- c) In order to recover the line for the humanities group, we require the intercept, the coefficient for final grade, the coefficient for Group3, and the interaction between final grade and Group3:

$$\hat{\beta}_0 = 1.9505, \quad \hat{\beta}_{\text{Group3}} = 3.1035, \quad \hat{\beta}_{\text{Group3}:\text{x}} = 2.9523.$$

To recover the y-axis (in the plot!) intercept of the humanities-degree line, we set the final grade equals to $x = 1.0$ and add the values of the coefficients (here it is assumed that there are no better grades than 1.0), i.e. [0.5P.]

$$\begin{aligned} \text{y-axis intercept [at } x = 1] &= \hat{\beta}_0 + \hat{\beta}_1 \cdot 1.0 + \hat{\beta}_{\text{Group3}} \cdot 1 + \hat{\beta}_{\text{Group3}:\text{x}} \cdot 1.0 \\ &= 1.9505 + 5.0195 + 3.1035 + 2.9523 \\ &= 13.0258 \quad [1.5P.] \end{aligned}$$

The slope of the line is given by the sum of the coefficients for the interaction, and final grade, i.e.

$$\begin{aligned} \text{slope} &= \hat{\beta}_1 + \hat{\beta}_{\text{Group3}:\text{x}} \\ &= 5.0195 + 2.9523 \\ &= 7.9718 \quad [1.5P.] \end{aligned}$$

The regression line for students with a humanities degree (Group3) would hence be given by:

$$\hat{y}_{\text{Group3},i} = 13.0258 + 7.9718 \cdot (x_i - 1), \quad [0.5P.]$$

or, more simple, as

$$\hat{y}_{\text{Group3},i} = 5.054 + 7.9718 \cdot x_i. \quad [0.5P.]$$

- d) Interpretation:

- For students with a degree in natural sciences (Group1) with a final grade of 1.0, the time between finishing their studies and their first employment is around 6 weeks, on average. For each **decimal point** of their final grade (i.e. a worse grade by one decimal point), their time until the first employment will increase by approx. 0.502 weeks on average. [2P.]
- For students with a degree in engineering (Group2) with a final grade of 1.0, the time between finishing their studies and their first employment is around 2.5 weeks, on average. For each **decimal point** of their final grade (i.e. a worse grade by one decimal point), their time until the first employment will increase by approx. 0.201 weeks ($\hat{\beta}_1 + \hat{\beta}_{\text{Group2}:\text{x}}$) on average. [2P.]
- For students with a degree in humanities (Group3) with a final grade of 1.0, the time between finishing their studies and their first employment is around 12.5 weeks, on average. For each **decimal point** of their final grade (i.e. a worse grade by one decimal point), their time until the first employment will increase by approx. 0.797 weeks ($\hat{\beta}_1 + \hat{\beta}_{\text{Group3}:\text{x}}$) on average. [2P.]

- e) The variable final grade exhibits the largest influence on the students with a humanities degree (Group3). [2P.]

(This can be seen directly in the plot, where the regression line for Group3 students has the steepest slope. This can also be concluded from the values of the estimated regression parameters. The value of the parameters for Group3 have larger and positive values compared to those of Group2.)

- f) The predictions for Group1 (naturalists) are :

$$\begin{aligned} \hat{y}_{\text{Group1}} &= \hat{\beta}_0 + \hat{\beta}_1 \cdot 2.0 \\ &= 1.9505 + 5.0195 \cdot 2.0 \\ &= 6.97 \quad [1P.] \end{aligned}$$

Predictions for Group2 (engineers) are:

$$\begin{aligned}\hat{y}_{\text{Group2}} &= \hat{\beta}_0 + \hat{\beta}_1 \cdot 2.0 + \hat{\beta}_{\text{Group2}} + \hat{\beta}_{\text{Group2:x}} \cdot 2.0 \\ &= 1.9505 + 5.0195 \cdot 2.0 + (-1.1278) + (-3.0054) \cdot 2.0 \\ &= 16.8725 \quad \text{[1P.]}\end{aligned}$$

Predictions for Group3 (humanities) are:

$$\begin{aligned}\hat{y}_{\text{Group3}} &= \hat{\beta}_0 + \hat{\beta}_1 \cdot 2.0 + \hat{\beta}_{\text{Group3}} + \hat{\beta}_{\text{Group3:x}} \cdot 2.0 \\ &= 1.9505 + 5.0195 \cdot 2.0 + 3.1035 + 2.9523 \cdot 2.0 \\ &= 20.9976 \quad \text{[1P.]}\end{aligned}$$